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DONGU KIM

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EDUCATION

Ulsan National Institute of Science and Technology (UNIST) 2022 - 2024
M.S. in Computer Science and Engineering

Inha University 2015 - 2022
B.S. in Electronic Engineering

RESEARCH INTERESTS

- Distributed System
- Hardware-software Co-design
- AI Infrastructure
- Programmable Network Devices

PUBLICATIONS

1. Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, Youngwoo Jeong, **Dongu Kim**, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim, "BoostX^{TM}-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration," *ISCA '26 (Industry Track)*, 2026.
2. **Dongu Kim**, Seyeong Park, Seyeon Kim, Sangtae Ha, Youngbin Im, "FlexServer: A High-Performance Content Server Leveraging Asynchronous I/O and SmartNICs," *Preprint*, 2024.

RESEARCH & PROFESSIONAL EXPERIENCE

Mangoboost Inc. Apr 2024 - Present
System Software Engineer

· **NVMe-over-TCP Initiator (NTI): Hardware-offloaded Virtualized NVMe/TCP Accelerator** *(flagship product; published at ISCA '26 Industry Track)*

- NVMe/TCP Initiator (NTI) software stack, with an ARM-FPGA-accelerated implementation that offloads NVMe/TCP administrative commands to onboard ARM cores and I/O commands to FPGA logic
  - ✓ Designed and implemented an SPDK-based userspace driver running on onboard Arm cores to process NVMe administrative commands, enabling compatibility across diverse host systems and contributing to the accelerator's UNH-IOL certification
  - ✓ Added system-level robustness and resilience mechanisms to handle datacenter failures, including network faults and NVMe/TCP target errors

· **MASS: Distributed Storage Infrastructure for AI**

- RDMA-based high-performance distributed storage system using DPU-JBoFs and flash arrays for efficient data delivery to GPUs and data sharing across GPUs
  - ✓ Develop a Kubernetes-based storage cluster management system supporting diverse storage backends, with storage server telemetry, hardware resource management including NICs and NVMe devices, and hardware/software deployment capabilities that enable seamless rolling upgrades for users
  - ✓ Integrate in-house FPGA-based RDMA NICs and DPU-based JBOF cards into the system, develop and integrate GPU-Direct RDMA (GDR) support for AMD GPUs into the system, and optimize and evaluate storage performance across diverse GPU workloads

Next Generation Networks and Systems Lab (*Advisor: Youngbin Im*) Mar 2022 - Feb 2024
Research Assistant

· **FlexServer: DPU-Accelerated Content Delivery**

- DPU-based content server that places files across host DRAM, DPU DRAM, and storage and overlaps TCP processing with asynchronous file I/O to hide its latency
  - ✓ Designed and implemented a multi-tier cache spanning DPU memory, host DRAM, and storage, using prefetching and object promotion/demotion to reduce accesses to slower storage
  - ✓ Overlapped asynchronous file reads with TCP processing to hide file I/O latency by processing TCP traffic for already available data while waiting for file reads to complete

PATENTS

1. Youngbin Im and Dong U Kim, "METHOD AND APPARATUS FOR IMPROVING KEY VALUE STORE PERFORMANCE THROUGH OFFLOADING OF SMART NETWORK INTERFACE CARD," KR Patent Registered No. 10-2724-6120000, Oct. 2024

HONORS & AWARDS

- Honor Student, Inha University Fall 2021
- Honor Student, Inha University Spring 2021

TEACHING EXPERIENCE

- Teaching Assistant, Computer Architecture (CSE261), UNIST Fall 2022
- Teaching Assistant, System Programming (CSE251), UNIST Spring 2022

TECHNICAL SKILLS

- **Languages**: C/C++, Go, Python, P4
- **Systems & Frameworks**: DPDK, SPDK, Kubernetes, DAOS, Mooncake, ROCm
- **Core Technologies**: RDMA, TCP, PCIe, NVMe, NVMe-oF
- **Hardware Platforms**: NVIDIA BlueField, Netronome Agilio SmartNIC, Xilinx FPGA, AMD GPU

EXTRACURRICULAR ACTIVITIES

Competitions & Projects

· **The World Embedded Software Contest 2019** *Apr 2019 - Dec 2019*

- Developed multithreaded software for an autonomous RC car featuring real-time obstacle avoidance; gained in-depth expertise in concurrent programming and thread synchronization
- Integrated various sensors—including LiDAR and Ultrasonic—with the software architecture to improve the accuracy of environmental mapping

Volunteer Work

· **Freview** *Feb 2020 - Jan 2021*

- Served as a volunteer mentor, sharing specialized knowledge about my major to help middle and high school students explore potential career paths and gain a deeper understanding of the field